

Exam-Paper-COMP2007J-Resit-Questions

COMP2007J Principles of Computer Organization — Resit Examination

Semester 2 Resit, 2024/2025. School of Computer Science, Beijing-Dublin International College.

Time allowed: 120 minutes.

Instructions for Invigilators

Candidates are allowed to use a non-programmable calculator. Any use of other devices or text materials during this examination is prohibited.

Question 1 (20 points)

A research team is evaluating the performance of a benchmark running on a new experimental processor. During testing, they observed the following characteristics:

- The benchmark executed 1.800×10^{12} instructions.
- The measured CPU execution time was 600 seconds.
- The processor has a clock cycle time of 0.25 ns.
- For comparison purposes, the team also recorded a reference execution time of 12,000 seconds from a baseline system.

The engineers are now analyzing how changes in the instruction set and microarchitecture might affect performance. Answer the following questions, showing your calculations clearly.

- (a) Using the performance equation, determine the CPI (Cycles Per Instruction) for this benchmark. (7 points)
- (b) Suppose the compiler optimizations change slightly, and the instruction count for this benchmark increases by 10%, while the CPI and clock cycle time remain unchanged. Calculate the new CPU execution time and state how much longer it is compared to the original run. (7 points)
- (c) Now assume that both the instruction count increases by 10% and the CPI increases by 5% due to additional pipeline stalls, while the clock cycle time remains unchanged. Compute the new CPU execution time and explain how the two factors (instruction count and CPI) combine to affect performance. (6 points)

Question 2 (20 points)

Translate the following C code to RISC-V assembly language. Assume that the values of `m`, `n`, `i`, and `j` are in registers `x5`, `x6`, `x7`, and `x28`, respectively. Also, assume that register `x11` holds the base address of the array `B`.

```
for (i = 0; i < m; i++)
    for (j = 0; j < n; j++)
        B[4*i + j] = i - j;
```

Question 3 (20 points)

In the process of executing programs written in high-level languages:

- (a) What are the main tasks of a compiler? Describe the steps it performs. (10 points)
- (b) How does an interpreter differ from a compiler? Describe its steps and when it is typically used. (10 points)

Question 4 (20 points)

Add NOP instructions to the code below so that it will run correctly on a pipeline that does not handle data hazards.

```
addi x5, x6, 8
sub  x7, x5, x6
addi x8, x5, 12
and  x9, x7, x5
```

Question 5 (20 points)

A CPU designer is considering the addition of a dedicated division unit. This enhancement adds 200 ps to the ALU latency but reduces the instruction count by 12% (since division no longer needs to be emulated).

Assume:

- $T = 2000 \text{ ps}$: original ALU latency (clock cycle time before improvement).
- $\Delta T = 200 \text{ ps}$: additional latency introduced by the new divider unit.
- I : original instruction count.
- $r = 0.12$: fraction reduction in instruction count.
- CPI : average cycles per instruction (assumed constant).

(a) What is the clock cycle time with and without this improvement? (6 points)

(b) What is the speedup achieved by adding this improvement? (7 points)

(c) What is the slowest new ALU that can be used and still result in improved performance? (7 points)

Question 6 (20 points)

Assume a 2-way set-associative cache with four one-word blocks, organized into 2 sets (i.e., each set can hold 2 blocks). Each memory address refers to a one-word block (i.e., the memory is word-addressable). Assume the cache is initially empty. Consider the following memory address sequence:

```
0, 1, 2, 3, 0, 2, 4, 5, 6, 4, 7, 5, 8, 9, 6, 7, 8, 9, 0, 2
```

(a) Assuming an LRU (Least Recently Used) replacement policy, which memory accesses are hits? (10 points)

(b) Assuming an optimal replacement policy for this sequence, which memory accesses are hits? (10 points)

Good luck.